Case Report

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Congenital Generalized Hypertrichosis, Gingival Hyperplasia, a Coarse Facies with Constriction Bands: A Rare Association

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ABSTRACT

Congenital generalized hypertrichosis terminalis is a rare primary hypertrichotic condition, of unknown etiology presenting in the pediatric population. Though benign in nature, there is considerable psychosocial trauma attached to this, owing to the cosmetic disfigurement it produces. The association of gingival fibromatosis and a coarse facies could further worsen the cosmesis. Thus, a multidisciplinary approach involving a psychologist, a dentist apart from the dermatologist would be mandatory. We present this rare syndrome with the purpose of getting a better insight regarding the inheritance, the clinical features and the best available treatment modalities, especially the modern and novel techniques of hair removal that could be utilized to manage such individuals.

Key words: Coarse facies, constriction bands, gingival hyperplasia, hypertrichosis

INTRODUCTION

Hypertrichosis is a clinical condition characterized by excessive hair growth without androgenic stimulation.[1] If there is no underlying cause of the excessive hair growth, it is referred to as primary hypertrichosis, which is further classified as congenital or acquired, based on the age of onset and generalized or localized based on the distribution of body hair.[2] Though not a serious condition clinically, the cosmetic burden on the child could be immense. Therefore, the psychological aspect of this disorder should not be neglected apart from instituting appropriate medical and esthetic therapies.

CASE REPORT

A 2-year-old girl presented to the Department of Dermatology with coarse hair present over the entire body since birth. She was born out of a non-consanguineous marriage with an uneventful antenatal and post-natal history, as described by the mother. She also had a 45-day-old sister who did not display this hairy phenotype. There were no associated systemic complications in the child. There was no history of photosensitivity. She was playful like any other normal 2-year-old and responded well to all commands. The main reason for the consult was the issue of cosmesis that alarmed the parents. On examination, terminal hair had a generalized distribution over the entire body [Figures 1 and 2], sparing the palms, soles, and the mucosal surfaces. The eyebrows were thick and bushy with an appreciable length of the eyelashes [Figure 3]. The helix of both ears were also coated with hair. The patient had a characteristic coarse facies with low set slightly enlarged ears. Bilaterally the arms demonstrated prominent constriction bands with noteworthy palmar dermatoglyphics [Figure 4]. Constriction bands were also seen in the upper portion of the medial aspect of both thighs [Figure 5]. Her oral cavity showed pink, hyperplastic and stippled gingiva with an altered dental architecture [Figure 6]. A genetic mapping study was planned for the same, but owing to financial constraints it was deferred. Patient was referred to the department of Dentistry for the gingival alteration and her parents were counseled regarding the benign nature of the disease with the possible options of available hair removal techniques, as the child grew older, for a better cosmetic outcome. Patient was asked for a regular review. However, the patient was subsequently lost to follow-up.



Figure 1: Terminal hair, seen over the face, chest, abdomen and the upper extremities. Also to note here is the coarse facies with low set slightly enlarged ears and constriction bands over the arms



Figure 2: Terminal hair over the back



Figure 3: Bushy eye-brows as seen in our patient



Figure 4: Prominent palmar dermatoglyphics



Figure 5: Terminal hair seen over bilateral lower limbs with incomplete constriction bands over the medial aspect of both the thighs



Figure 6: Hyperplastic stippled gingival architecture with dental disarray

DISCUSSION

Congenital generalized hypertrichosis (CGH) represents a heterogeneous group of conditions that are phenotypically and genetically distinct, and is characterized by excessive universal hair growth as the hallmark feature, that is disproportionate when compared to normal individuals of that age, sex and race and is not dependent on androgenic

hormones.[3] CGH has been seen as the major phenotype in 10 different genetic syndromes. These include:

- Congenital hypertrichosis lanuginosa (CHL)
- Ambras syndrome (AS)
- Congenital generalized hypertrichosis terminalis with gingival hyperplasia
- X-linked dominant form of hypertrichosis
- Barber Say syndrome (BSS)
- Cantu syndrome (CS)
- Amaurosis congenita, cone-rod type with congenital hypertrichosis
- Cataract, hypertrichosis, and mental retardation syndrome
- Gingival fibromatosis with hypertrichosis and mental retardation
- Other genetic disorders with congenital hypertrichosis, e.g. Zimmermann-Laband syndrome, Coffin-Siris syndrome (CSS) and others.

Congenital hypertrichosis lanuginosa

Congenital hypertrichosis lanuginosa is a very rare hypertrichotic disorder with an autosomal dominant (AD) inheritance pattern although sporadic presentations have been reported.[4] Normally at birth the lanugo hair is shed and replaced by vellus hair. However, in this clinical setting the lanugo hair persists.[5] Lanugo hair is a special, downy, non-medullated and non-pigmented hair that covers the entire foetus upto 6 months of gestation, following which it is shed. Lanugo hair has a smooth surface with almost indiscriminate scales. These hairs may reach a length of 10 cm and blend with the darker terminal hair over the scalp and the eyebrows. Terminal hair, on the other hand, is thick and pigmented and is usually seen following androgenic stimulation. However, in CGH terminalis this hormonal stimulation may not be associated. Vellus hair is a soft, fine, unmedulated hair, which rarely exceeds 2 cm in length. A knowledge of the various hair types is, therefore, essential while diagnosis these hypertrichotic syndromes. A characteristic feature of CHL is the striking abundance of hair in the dorsal and lumbosacral spine with a characteristic whirling pattern in the sacral and pre-auricular regions.[6,7] This disorder may have other congenital defects associated with it, namely pyloric stenosis, Fallot's tetralogy, congenital glaucoma and growth retardation.[8]

Ambras syndrome

Ambras syndrome is a rare congenital hypertrichotic syndrome characterized by long, fine, vellus hair covering the entire body, sparing the palms, soles, mucous membranes and the dorsal terminal phalanges.[9] Along with this, there is associated coarse facies, multiple exostosis, postaxial rudimentary hexadactyly, teeth abnormalities and a low insertion of the first metacarpal.[10,11] An AD pattern of inheritance has been proposed for AS. A position defect in the TRPS1 gene within the 8q22-8q24 chromosomal region has been found to have a causative role in AS.[12]

Congenital generalized hypertrichosis terminalis with gingival hyperplasia

This is a distinct clinical entity with growth of fully pigmented terminal hair over the entire body, sparing the palms, soles, and the mucosa. Along with the hypertrichosis, the phenotype comprises gingival hypertrophy and a coarse facies.[13,14] It is a very rare condition with an AD pattern of inheritance.[15] It has been associated with a recurrent microdeletion on the 17q24.2-17q24.3 chromosome loci, and this disorder has now been clearly established as a genomic disorder.[16] Our patient too, presented with this phenotype. However along with the coarse facies and gingival hyperplasia incomplete constriction bands were identified over the upper extremities and the medial aspect of bilateral thighs. This was a new finding witnessed in our patient, and to the best of our knowledge and review of literature, the association of constriction bands in this scenario has not been reported till date. Genetic mapping though contemplated in our case was not performed owing to financial constraints.

X-linked dominant hypertrichosis

This is an X-linked dominant disorder characterized by excessive terminal hair growth over the entire body sparing the palms, soles and the mucosal surfaces, without gingival hypertrophy,[17,18] to distinguish it from the previous entity. Figuera *et al*.[19] elucidated that the involved region in the X chromosome belonged to the interval between the DXS425 and the DXS1227 regions in the Xq24-Xq27.1 chromosome. This finding was further substantiated by Zhu *et al*.[20]

Barber Say syndrome

Barber Say syndrome is an AD syndrome with a constellation of clinical features.[21] Some of the features include disproportionate terminal hair growth over the forehead and back, a coarse facial look with hypertelorism, bilateral ectropion, macrostomia and a bulbous nose. The skin is lax and atrophic mimicking that of an elderly individual, and there is failure to thrive.[22,23]

Cantu syndrome

Cantu syndrome is a rare hypertrichotic syndrome with an AD pattern of inheritance.[24,25] Clinically apart from

hypertrichosis, osteochondrodysplasia and cardiomegaly are the hallmark features. To note here also is the coarse facial features which consists of a wide mouth, broad nasal bridge, epicanthal folds and full lips,[26] which needs a thorough work-up to correctly distinguish it from other hypertrichotic syndromes.

Amaurosis congenita, cone-rod type with congenital hypertrichosis

This is an autosomal recessive (AR) hypertrichotic syndrome.[27] The cardinal features displayed here apart from the hypertrichotic phenotype include severe visual impairment secondary to retinal dystrophy along with marked photophobia in the absence of night blindness.

Cataract, hypertrichosis, and mental retardation syndrome

This is another inherited syndrome associated with hypertrichosis. It has an AR pattern of inheritance. The clinical findings seen here include congenital lamellar cataracts, generalized hypertrichosis involving the shoulders, face and back along with mental retardation.[28]

Gingival fibromatosis with hypertrichosis and mental retardation

In this condition, there is generalized hypertrichosis of the terminal hair type associated with coarse facies, gingival hyperplasia, epilepsy and mental retardation. Apart from these hypothyroidism and congenital cardiomyopathy are associated features.[29,30]

Zimmermann-Laband syndrome

This is an AD condition and may present with variable phenotypes.[31,32] Clinically it is characterized by gingival hyperplasia, abnormalities involving the face mainly the nose and the ears with joint hyperextensibility and hypoplasia confined to the nails and the terminal phalanges of the hands and feet. Apart from these findings generalized hypertrichosis, and mental retardation are other features to take note of.[33]

Coffin-Siris syndrome

Coffin-Siris syndrome is an AR disorder[34] with studies showing that 7q32-q34 region contains the gene that eventually bears the onus for CSS.[35] Clinically apart from hypertrichosis and a coarse facies, CSS is characterized by absence of the fifth finger nails and toe nails, underdeveloped distal phalanges of the little fingers and the second to the fifth toes, small patellas, sucking and feeding difficulties and inguinal hernia.[36] A syndrome, which may show overlapping with CSS is the Cornelia de Lange syndrome (CDLS) and must be kept in mind. CDLS is characterized by hypertrichosis, mental retardation, microcephaly and a peculiar facies.[37]

Hence, we see a number of syndromes with hypertrichosis as the hallmark feature. There are a number of more syndromes encountered here, but discussing all would be beyond the scope of this article. As these disorders are associated with a multitude of systemic involvement, a multidisciplinary approach would be required for managing these patients, involving the concerned speciality in the treatment. As far as the management of hypertrichosis is concerned, long-term removal of hair in these patients poses to be a challenging concern. This in turn is dependent upon the degree of hair growth, the patient's psychologic profile and the issue of social acceptance. Epilatory methods of hair removal, whereby the entire hair shaft is uprooted,[38] would be a better method here than the depilatory methods because of the longer duration of hair free skin got by it. Epilatory methods include mechanical or electronic tweezers,[39] wax epilation or the use of a twisted string run rapidly over a hair bearing area epilating the hair thus.[40] These methods maybe of use in small areas of the body, for example, the face as they are painful and may not receive the full co-operation of the child. The other newer therapies using lasers and lights may be of help for removing extensive areas of excessive hair growth, where the principle of thermolysis is applied to selectively target the melanin rich hair follicles, which are thus destroyed causing minimal surrounding tissue affection. However, their use in pre-pubertal children has not been extensively studied. Morley *et al*.[41] have assessed the use of the long-pulsed ruby laser in pre-pubertal children and have concluded that it is a safe and effective technique for epilation with pertinent levels of patient and parent satisfaction. According to Littler[42] the Q-switched Nd: Yag laser at low fluences was effective in markedly reducing the hair density in a patient of hypertrichosis lanuginosa congenita. As these disorders are so rare large studies in this aspect is lacking. A novel therapy, using eflornithine hydrochloride, an ornithine decarboxylase inhibitor,[43] available as a 13.9% cream has shown to hold promise in the issue of hair growth retardation in women. However, in the pediatric population the efficacy and safety for the same has not been established. To conclude hypertrichosis on its own is still an enigma, which needs further research in order to get a vivid understanding in this context. Secondly as far as therapy is concerned, a better knowledge of the clinico-pathogenesis could tremendously prove useful in this setting for the exploration of newer modalities which could definitely help to outline a better quality of life in these patients. With these points in mind, we present this case.

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